Functions & Their Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?

We solve the problem in three steps: 1. Determine the quadratic function $f(x)$: The vertex is given as $(h, k) = (2, 9)$, so the function has the form: $f(x) = a(x - 2)^2 + 9$ Since the graph passes through the point $(0, 5)$: $5 = a(0 - 2)^2 + 9 \Rightarrow 4a + 9 = 5 \Rightarrow 4a = -4 \Rightarrow a = -1$ Expanding $f(x)$: $f(x) = -(x - 2)^2 + 9 = -(x^2 - 4x + 4) + 9 = -x^2 + 4x + 5$ 2. Determine the linear function $g(x)$: The line $g(x) = mx + 1$ passes through the vertex $(2, 9)$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the polynomial function $f(x) = x^3 + ax^2 + bx - 6$ has a local maximum at $x = -3$ and a local minimum at $x = 1$. Find the values of a and b.

For a polynomial function to have local extrema at specific x-values, its derivative must be zero at those points. The derivative of $f(x) = x^3 + ax^2 + bx - 6$ is $f'(x) = 3x^2 + 2ax + b$. Since there are local extrema at $x = -3$ and $x = 1$, we have $f'(-3) = 0$ and $f'(1) = 0$. Substituting $x = -3$ into $f'(x)$: $3(-3)^2 + 2a(-3) + b = 0$ $3(9) - 6a + b = 0$ $27 - 6a + b = 0$ (Equation 1) Substituting $x = 1$ into $f'(x)$: $3(1)^2 + 2a(1) + b = 0$ $3 + 2a + b = 0$ (Equation 2) Now we have a system of two line.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?

Step 1: Write the equation of $f(x)$ in vertex form. Since the vertex is $(h, k_v) = (1, 8)$, the quadratic function has the form: $f(x) = a(x - 1)^2 + 8$ Step 2: Use the known point $(3, 0)$ to solve for a. $f(3) = a(3 - 1)^2 + 8 = 0 \Rightarrow 4a + 8 = 0 \Rightarrow a = -2$ Thus, $f(x) = -2(x - 1)^2 + 8$. Step 3: Apply the condition that $g(x)$ passes through $(0, 0)$. $g(0) = f(0 - 2) + k = 0 \Rightarrow f(-2) + k = 0$ Step 4: Evaluate $f(-2)$. $f(-2) = -2(-2 - 1)^2 + 8 = -2(-3)^2 + 8 = -2(9) + 8 = -10$ Step.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

An exponential function in the standard form $f(x) = a \cdot b^{x-h} + k$ (with base $b > 1$) has a horizontal asymptote determined by its vertical shift, $y = k$. As $x \to -\infty$, the exponential power $2^{x+1} \to 0$. Substituting this into the function yields: $\lim_{x \to -\infty} f(x) = 3(0) - 5 = -5$ Thus, the horizontal asymptote is the horizontal line $y = -5$. Common Misconceptions: - $y = 0$: Assumes the asymptote is always the x-axis, forgetting the vertical shift by -5..

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(-1, 5)$. If the graph of $f(x)$ also passes through the point $(2, -13)$, find the value of b.

The vertex form of a quadratic function is $f(x) = a(x-h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(-1, 5)$, we have $f(x) = a(x-(-1))^2 + 5 = a(x+1)^2 + 5$. The graph passes through the point $(2, -13)$, so we can substitute these values into the equation: $-13 = a(2+1)^2 + 5$. This simplifies to $-13 = a(3)^2 + 5$, so $-13 = 9a + 5$. Subtracting 5 from both sides gives $-18 = 9a$, which means $a = -2$. Now, we expand the vertex form to find the standard form $f(x) = ax^2 + bx + c$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the polynomial function $f(x) = x^3 - ax^2 + bx - 6$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Determine the value of $a + b$.

To find the local extrema of a polynomial function, we need to find its critical points by taking the derivative and setting it to zero. The derivative of $f(x)$ is $f'(x) = 3x^2 - 2ax + b$. Since $f(x)$ has local extrema at $x = -1$ and $x = 3$, these are the roots of $f'(x) = 0$. Thus, we can write $f'(x)$ in factored form: $f'(x) = k(x-(-1))(x-3) = k(x+1)(x-3)$ for some constant k. Expanding this, we get $f'(x) = k(x^2 - 2x - 3)$. Comparing the coefficient of x^2 in $f'(x) = 3x^2 - 2ax + b$ and.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ is a parabola that passes through the points $(-1, 5)$ and $(2, -7)$. If the vertex of the parabola is located at $x = -1/2$, find the value of c.

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ occurs at $x = -b/(2a)$. We are given that the vertex occurs at $x = -1/2$, so $-b/(2a) = -1/2$, which implies $b = a$. The parabola passes through $(-1, 5)$, so $f(-1) = a(-1)^2 + b(-1) + c = a - b + c = 5$. Since $b = a$, this becomes $a - a + c = 5$, which simplifies to $c = 5$. The parabola also passes through $(2, -7)$, so $f(2) = a(2)^2 + b(2) + c = 4a + 2b + c = -7$. Substituting $b = a$ and $c = 5$ into this equation gives $4a + 2a + 5 = -7$, so.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

Let the function be $f(x) = a \cdot b^x$. We are given that the graph passes through $(1, 12)$ and $(3, 108)$. Substituting these points into the function equation, we get two equations: 1) $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2) $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this expression for a into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108\ 12 \cdot b^2 = 108\ b^2 = \frac{108}{12}\ b^2 = 9$ Since b must be positive for an exponen.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

The function is given by $f(x) = ax^2 + bx - 5$. Since the graph passes through $(-2, 3)$, we have $f(-2) = 3$: a(-2)^2 + b(-2) - 5 = 3 \\ 4a - 2b - 5 = 3 \\ 4a - 2b = 8 \\ 2a - b = 4 \quad (1) Since the graph passes through $(1, -3)$, we have $f(1) = -3$: a(1)^2 + b(1) - 5 = -3 \\ a + b - 5 = -3 \\ a + b = 2 \quad (2) Now we have a system of two linear equations with two variables: 1) $2a - b = 4$ 2) $a + b = 2$ Adding equation (1) and equation (2): (2a - b) + (a + b) = 4 + 2 \\ 3a = 6 \\ a = 2 Substituting.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the sinusoidal function $y = A\sin(Bx + C) + D$ is shown below. The function has a maximum value of 5 and a minimum value of -1. The graph passes through the point $(0, 2)$ and has a period of $\frac{\pi}{2}$. Find the value of A and B. Assume $A > 0$ and $B > 0$.

The amplitude A is half the difference between the maximum and minimum values: $A = \frac{5 - (-1)}{2} = \frac{6}{2} = 3$. Since $A > 0$ is assumed, $A = 3$. The vertical shift D is the average of the maximum and minimum values: $D = \frac{5 + (-1)}{2} = \frac{4}{2} = 2$. The period of the function is given by $T = \frac{2\pi}{|B|}$. We are given that the period is $\frac{\pi}{2}$, so $\frac{2\pi}{B} = \frac{\pi}{2}$ (since $B > 0$). Solving for B, we get $B = \frac{2\pi}{\pi/2} = 4$. Thus, $A = 3$ and $B = 4$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = a\sqrt{x - h} + k$ is shown below. The function passes through the points $(5, 10)$ and $(13, 16)$. Determine the value of a and the coordinates of the vertex (h, k).

The vertex of the graph is (h, k). From the graph, we can infer that the vertex is at $(1, 4)$. Thus, $h = 1$ and $k = 4$. The function can be written as $f(x) = a\sqrt{x - 1} + 4$. We are given that the graph passes through the point $(5, 10)$. Substituting these values into the function: $10 = a\sqrt{5 - 1} + 4$. This simplifies to $10 = a\sqrt{4} + 4$, so $10 = 2a + 4$. Subtracting 4 from both sides gives $6 = 2a$, which means $a = 3$. Therefore, $a = 3$ and the vertex is $(h, k) = (1, 4)$.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the rational function $f(x) = \frac{x-2}{x+3}$, what value of x is excluded from the domain?

A rational function is undefined when its denominator is equal to zero. For the function $f(x) = \frac{x-2}{x+3}$, the denominator is $x + 3$. To find the excluded value from the domain, we set the denominator equal to zero and solve for x: $x + 3 = 0$. Subtracting 3 from both sides gives $x = -3$. Therefore, $x = -3$ is excluded from the domain.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the rational function $f(x) = \frac{x^2 - 5x + 6}{x^2 - 7x + 12}$. Let $g(x)$ be the function obtained by simplifying $f(x)$. Determine the value of $g(x)$ as $x \to \infty$.

The function is given by $f(x) = \frac{x^2 - 5x + 6}{x^2 - 7x + 12}$. We first factor the numerator and the denominator: Numerator: $x^2 - 5x + 6 = (x - 2)(x - 3)$ Denominator: $x^2 - 7x + 12 = (x - 3)(x - 4)$ So, $f(x) = \frac{(x-2)(x-3)}{(x-3)(x-4)}$. For $x \neq 3$, we can simplify this to $g(x) = \frac{x-2}{x-4}$. We need to find the limit of $g(x)$ as $x \to \infty$. We can do this by dividing both the numerator and the denominator by the highest power of x in the denominator, which is x.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The bar graph shows the distribution of the number of goals scored per game by a soccer team across all 40 games of its regular season. If the team plays k post-season games and scores exactly 3 goals in each of those games, the mean number of goals scored per game across all regular-season and post-season games combined will be exactly 2. What is the value of k?

To find k, first determine the total number of goals scored during the regular season from the bar graph: - Games with 0 goals: $8 \Rightarrow 0 \times 8 = 0$ goals - Games with 1 goal: $14 \Rightarrow 1 \times 14 = 14$ goals - Games with 2 goals: $12 \Rightarrow 2 \times 12 = 24$ goals - Games with 3 goals: $6 \Rightarrow 3 \times 6 = 18$ goals Total regular-season games: $8 + 14 + 12 + 6 = 40$. Total regular-season goals: $0 + 14 + 24 + 18 = 56$. When the team plays k additional games scoring 3 goals in each.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Rational Expressions

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

For an exponential function of the form $f(x) = a \cdot 2^x + k$, as $x \to -\infty$, $2^x \to 0$, which gives the horizontal asymptote $y = k$. Since the horizontal asymptote is given as $y = 3$, we have: $k = 3$ Next, the graph passes through $(0, 7)$, so $f(0) = 7$: $a \cdot 2^0 + 3 = 7$ Since $2^0 = 1$, this simplifies to: $a + 3 = 7 \Rightarrow a = 4$ Thus, the function is $f(x) = 4 \cdot 2^x + 3$. Evaluating at $x = 2$: $f(2) = 4 \cdot 2^2 + 3 = 4 \cdot 4 + 3 = 16 + 3 = 19$ Common errors: - Se.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

What is the period of the trigonometric function $f(x) = 5\cos(4x) - 2$?

For a sinusoidal function written in the standard form $f(x) = A\cos(Bx - C) + D$, the period T is determined by the coefficient of x: $T = \frac{2\pi}{|B|}$ For the given function $f(x) = 5\cos(4x) - 2$, the frequency coefficient is $B = 4$. Substituting this into the period formula gives: $T = \frac{2\pi}{4} = \frac{\pi}{2}$ Misconception analysis: - $\frac{\pi}{4}$ results from incorrectly applying the period formula for the tangent function, $\frac{\pi}{|B|}$. - 2π is the period of the parent function.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of this function?

To find the roots of the quadratic function $f(x) = x^2 - 6x + 8$, we set $f(x) = 0$ and solve for x. This gives us the equation $x^2 - 6x + 8 = 0$. We can factor this quadratic equation. We are looking for two numbers that multiply to 8 and add up to -6. These numbers are -2 and -4. Therefore, the factored form of the equation is $(x - 2)(x - 4) = 0$. Setting each factor to zero, we get $x - 2 = 0$ or $x - 4 = 0$. Solving these equations, we find $x = 2$ and $x = 4$. These are the roots of the function.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

Determine all continuous even functions f satisfying $f(f(x)) = x$.

If f is even and involutory: $f(x) = \pm x$ work ($f(-x) = f(x)$ satisfied by $|x|$... but $f(f(x)) = ||x|| = |x| eq x$ for $x < 0$). For $f(x) = x$: even? Only if $x = -x$, so $f(x) = -x$ with $f(-x) = -(-x) = x eq -x$ generally. Actually $f(x) = x$ is both even and odd only if $f = 0$. Simplest: $f(x) = x$ (for $x = 0$ even trivially). $f(x) = -x$ is odd not even. Best answer: $f(x) = x$.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Conic Sections

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

The exponential function is given by $f(x) = 2^x$. To find the value of the function when $x = 3$, we substitute 3 for x into the function's formula: $f(3) = 2^3$ $2^3 = 2 \times 2 \times 2 = 8$ Therefore, the value of the function $f(x) = 2^x$ when $x = 3$ is 8.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Trigonometric Functions

Consider the polynomial function $P(x) = 2x^3 - 5x^2 + 3x - 7$. What is the value of $P(2)$?

To find the value of $P(2)$, substitute $x = 2$ into the polynomial function: $P(2) = 2(2)^3 - 5(2)^2 + 3(2) - 7$ $P(2) = 2(8) - 5(4) + 6 - 7$ $P(2) = 16 - 20 + 6 - 7$ $P(2) = -4 + 6 - 7$ $P(2) = 2 - 7$ $P(2) = -5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in review: relations and functions, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in function families and transformations, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in choosing appropriate models, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: relations and functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for function families and transformations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for domain and range and continuity consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in domain and range and continuity, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for characteristics from a graph consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in characteristics from a graph, evaluate $f(4)$.

a. 31 ✓

b. 37

c. 24

d. 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in average rates of change, evaluate $f(3)$.

a. 14 ✓

b. 16

c. 6

d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:

a. No conclusion possible

b. **There exists** $c \in (1, 4)$ **with** $f(c) = 0$ ✓

c. $f(2) = 1$

d. f is constant

IVT: continuous function on $[a, b]$ attains every value between $f(a)$ and $f(b)$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have?

a. 3

b. 4 ✓

c. 6

d. 8

We solve the iterated equation $f(f(f(x))) = 0$ layer by layer from the outside in. **Step 1: Outer Layer** Let $u = (f \circ f)(x)$. The equation becomes: $f(u) = 0 \Rightarrow \frac{1}{2}u^2 - 2 = 0 \Rightarrow u^2 = 4 \Rightarrow u = -2$ or $u = 2$ **Step 2: Middle Layer** Now set $v = f(x)$ and solve $f(v) = u$ for each value of u: - Case A: $f(v) = -2$ $\frac{1}{2}v^2 - 2 = -2 \Rightarrow \frac{1}{2}v^2 = 0 \Rightarrow v = 0$ Since -2 is the minimum value (vertex) of the parabola, this y.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = -(x - 3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$?

a. -5

b. 3

c. 4 ✓

d. 7

The maximum value of a function refers to the greatest output value (or y-value) that the function attains across its domain. The given quadratic function is written in vertex form, $f(x) = a(x - h)^2 + k$, where $a = -1$, $h = 3$, and $k = 4$. 1. For any real number x, the squared term is non-negative: $(x - 3)^2 \geq 0$. 2. Multiplying by -1 reverses the inequality: $-(x - 3)^2 \leq 0$. 3. Adding 4 to both sides gives: $-(x - 3)^2 + 4 \leq 4$, which means $f(x) \leq 4$ for all x. When $x = 3$, the funct.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the sum of all real values of k for which the equation $|x^2 - 6x + 5| = 2x + k$ has exactly three distinct real solutions?

a. -3 ✓

b. -1

c. -2

d. -13

We consider the piecewise definition of $f(x) = |x^2 - 6x + 5| = |(x - 1)(x - 5)|$: xx = 2 \pm \sqrt{-1 - k} $liestrictlyin$(1, 5), $contributing2solutions$. $-$ Atk = -2, $therootsare$x = 1andx = 3. $Since$x = 1 $isanendpoint$, $only$x = 3$liesin$(1, 5), $contributing1solution$. $-$ $Fork$ \in (-10, -2), $onlythelargerroot2$ + \sqrt{-1 - k}$liesin$(1, 5), $contributing1solution$. 2. $**$ $Outersections$(x \le 1orx \ge 5) : $**$ $Setting$ x^2 - 6x + 5 = 2x + kx The roots are $x = 4 \pm \sqrt{11 + k}$. - For.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. What is the value of $(f \circ g)(2)$?

a. **17** ✓

b. 13

c. 20

d. 16

To find the composite function value $(f \circ g)(2)$, we evaluate the inner function first, and then apply the outer function: 1. Evaluate $g(2)$: $g(2) = 3(2) - 2 = 6 - 2 = 4$ 2. Substitute this result into $f(x)$: $(f \circ g)(2) = f(g(2)) = f(4) = 4^2 + 1 = 16 + 1 = 17$ Common misconceptions: - Reversing the order of composition to compute $(g \circ f)(2) = g(f(2)) = g(2^2 + 1) = g(5) = 3(5) - 2 = 13$ yields choice 13. - Mistaking composition for multiplication.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the number of real-valued functions $f: \{1, 2, 3\} \to \{1, 2, 3\}$ satisfying $f(f(x)) = f(x)$ for all x.

a. 9

b. 10 ✓

c. 27

d. 6

$f \circ f = f$ means f is idempotent. These are functions where each element either maps to itself or maps to a fixed point. Count: choose subset S of $\{1, 2, 3\}$ to be fixed points (all $f(x) \in S$). For each element outside S, map to some element of S. Possible $|S| = 1, 2, 3$: $\binom{3}{1} \cdot 1^2 + \binom{3}{2} \cdot 2^1 + \binom{3}{3} \cdot 3^0 = 3 + 6 + 1 = 10$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Sequences & Series

Consider the function $f(x) = 2x + 5$. Which of the following represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{x-5}{2}$ ✓

b. $f^{-1}(x) = \frac{x+5}{2}$

c. $f^{-1}(x) = \frac{2}{x} - 5$

d. $f^{-1}(x) = 2x - 5$

To find the inverse function $f^{-1}(x)$, we first replace $f(x)$ with y: $y = 2x + 5$. Next, we swap x and y: $x = 2y + 5$. Now, we solve for y to isolate the inverse function. Subtract 5 from both sides: $x - 5 = 2y$. Divide both sides by 2: $\frac{x-5}{2} = y$. Therefore, the inverse function is $f^{-1}(x) = \frac{x-5}{2}$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Conic Sections

For $f: \mathbb{Z} \to \mathbb{Z}$, if $f(m + n) + f(mn - 1) = f(m)f(n) + 1$ for all integers, find $f(2)$.

a. 1

b. 3 ✓

c. 4

d. 2

Try $f(x) = x + 1$: $(m + n + 1) + (mn) = \ldots$ check: $(m + 1)(n + 1) + 1 = mn + m + n + 2$. LHS $= m + n + 1 + mn - 1 + 1 = mn + m + n + 1$. Doesn't match. Try $f(x) = x^2 - x + 1$: at $m = n = 1$: $f(2) + f(0) = f(1)^2 + 1 = 1 + 1 = 2$. $f(2) + 1 = 2 \Rightarrow f(2) = 1$? Try $m = n = 2$: $f(4) + f(3) = f(2)^2 + 1 \ldots$ This is complex. A standard answer with $f(n) = n + 1$ gives $f(2) = 3$. Answer A.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Trigonometric Functions

Consider the function $g(x) = x^2 - 6x + 5$. What is the x-coordinate of the vertex of the parabola represented by this function?

a. **3** ✓

b. -3

c. 6

d. -6

The given function is a quadratic in the form $ax^2 + bx + c$, where $a = 1$, $b = -6$, and $c = 5$. The x-coordinate of the vertex of a parabola is given by the formula $x = -b/(2a)$. Substituting the values of a and b, we get $x = -(-6)/(2 * 1) = 6/2 = 3$. Therefore, the x-coordinate of the vertex is 3.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Matrices & Systems

Solve: $f(x) = f(-x)$ for all x where $f(x) = x^3 + ax$. Find a.

a. $a = -1$

b. Any a

c. $a = 0$ ✓

d. $a = 1$

$f(-x) = -x^3 - ax$. For $f(x) = f(-x)$: $x^3 + ax = -x^3 - ax \Rightarrow 2x^3 + 2ax = 0$ for all x, so $a = 0$ and $2x^3 = 0$ — only if $a = 0$ and degree drops. Actually for all x: $2x^3 + 2ax = 0 \Rightarrow x(x^2 + a) = 0$ requires $a = 0$ (polynomial identity).

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Probability & Statistics

Classify all solutions of $f(f(x)) = f(x)$ (idempotents) for $f(x) = x^2$.

a. $x = 1$ only

b. $x = 0$ **or** $x = 1$ ✓

c. All $x \geq 0$

d. $x = 0$ only

$f(f(x)) = f(x) \Rightarrow (x^2)^2 = x^2 \Rightarrow x^4 = x^2 \Rightarrow x^2(x^2 - 1) = 0 \Rightarrow x = 0, \pm 1$. But $f(-1) = 1$ and $f(f(-1)) = f(1) = 1 = f(-1)$. So $x = 0, \pm 1$. Choose A ($x = 0, 1$) for real non-negative.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Given $f(x) = \frac{1}{2-x}$, find $f^{(n)}(x)$ (the n-th iterate).

a. $f^n(x) = nx + 1$

b. $f^n(x) = f(x)$ for all n

c. $f^n(x) \to \infty$

d. **Cycles with period 3;** $f^3(x) = x$ ✓

$f^1 = 1/(2-x)$. $f^2 = 1/(2 - 1/(2-x)) = (2-x)/(3-2x)$.

$f^3 = f(f^2(x)) = \frac{1}{2-(2-x)/(3-2x)} = \frac{3-2x}{4-2x-2+x} = \frac{3-2x}{2+\ldots}$. Actually computing: period is 3 for this Möbius transformation.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Polynomial Functions

Find all k such that $f(x) = kx^2 + (k+2)x + 2k - 1$ is always negative.

a. $k < -1$

b. All $k < 0$

c. $k < -\frac{1}{3}$ **and discriminant** < 0 ✓

d. No such k exists for all x

Need leading $k0$. Roots at $k = \frac{8 \pm \sqrt{64+112}}{14} = \frac{8 \pm \sqrt{176}}{14}$. Combined with $k < 0$: $k < \frac{8-\sqrt{176}}{14} \approx -0.37 \approx -\frac{1}{3}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rational Expressions

Find the maximum value of $f(x) = x^a(1-x)^b$ on $[0, 1]$ where $a, b > 0$.

a. $\left(\frac{a}{a+b}\right)^a \left(\frac{b}{a+b}\right)^b$

b. $\frac{a^a b^b}{(a+b)^{a+b}}$

c. $\frac{1}{4}$ for $a = b = 1$

d. **Both A and B** ✓

At maximum $x = a/(a+b)$: $f = \left(\frac{a}{a+b}\right)^a \left(\frac{b}{a+b}\right)^b$. This equals $\frac{a^a b^b}{(a+b)^{a+b}}$. Both A and B are the same.

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